

# SMARTBRIDGE / SKILL WALLET

## Technology Stack Blueprint

|                                                                      |            |
|----------------------------------------------------------------------|------------|
| Date: 2 July 2026                                                    | Team ID: — |
| Project Name: Caption AI : AI Powered Social Media Caption Generator |            |

### Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

### Technology Stack Details

| S.No | Architecture Component/Layer | Technology Chosen                          | Justification / Purpose                                                                                                                                                       |
|------|------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side       | HTML5, CSS3, JavaScript                    | Provides a lightweight, highly responsive user interface without heavy client-side rendering engine overhead, ensuring fast rendering across mobile and desktop environments. |
| 2    | Backend / Server-Side        | Python (Flask)                             | Chosen for its minimalist overhead, high speed in routing JSON server payloads, and native out-of-the-box support for AI library client integrations.                         |
| 3    | Database / Data Storage      | Environment Variable Configurations (.env) | Utilizes localized dot-env tracking to cleanly manage configurations and isolate API keys safely without structural database connection dependencies.                         |
| 4    | Cloud / Hosting / Deployment | Local Micro-server Agent                   | Maintains an isolated local server instance during rapid development cycles to avoid early production hosting overhead while remaining completely cost-effective.             |
| 5    | Version Control & CI/CD      | GitHub                                     | Enables secure file state tracking, version staging control, and clear backups across individual programming sprints.                                                         |

| S.No | Architecture<br>Component/Layer           | Technology Chosen                     | Justification / Purpose                                                                                                                                              |
|------|-------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | <b>Third-Party APIs / Other<br/>Tools</b> | Groq API Core & Ngrok<br>Core Proxies | Groq API executes the lightning-fast natural<br>language generation model cycles, while Ngrok<br>safely bridges local port boundaries to provide<br>clean endpoints. |